M-Series Motors: Maintenance and Troubleshooting

1. Scope

This guide describes preventive maintenance and troubleshooting for M-Series TEFC induction motors. Ratings and operating limits are given in M-OM-03. Insulation and winding tests are in M-ETP-05 (restricted to qualified electrical technicians).

All work on the motor or the terminal box requires isolation according to SOP-LOTO-001. Check for absence of voltage before opening the terminal box.

2. Preventive maintenance schedule

Interval	Task
Monthly	Clean the cooling fins and the fan cover inlet. Check for unusual noise.
Every 3 months	Check winding temperature trend and running current against nameplate.
Per lubrication table (M-OM-03 section 5)	Regrease the bearings.
Every 12 months	Measure insulation resistance (M-ETP-05). Check terminal connections for tightness and discolouration.
Every 12 months	Check alignment and vibration on both bearing housings.
Every 40,000 hours	Replace both bearings.

2.1 Cleaning the cooling path

TEFC motors depend on the air flow from the shaft-mounted fan over the finned frame. Dust, fibres or oil mist on the fins act as insulation and can raise the winding temperature by 10 to 20 °C. Clean the fins with a brush or dry compressed air at a maximum of 2 bar, with the motor isolated. Never use a water jet on the fan cover.

3. Fault table

Symptom	Probable cause	Remedy
Overheating	Blocked or dirty cooling fins / fan cover	Clean the cooling path (section 2.1).
Overheating	Damaged or missing cooling fan	Replace the fan; check fan cover clearance.
Overheating	Mechanical overload of the driven machine	Measure current; check the driven load (jammed conveyor, worn bearings).
Overheating	Supply voltage imbalance above 1%	Measure phase voltages; correct the supply.
Overheating	Single phasing (loss of one phase)	Check fuses, contactor contacts and terminals.
Overheating	Too many starts per hour	Limit to 3 starts per hour.
Overheating	High ambient temperature (above 40 °C)	Improve ventilation or derate.
High vibration	Misalignment or soft foot	Re-align; correct soft foot to 0.05 mm.
High vibration	Bearing damage	Replace the bearings.
High vibration	Rotor imbalance or broken rotor bar	Balance the rotor; check current spectrum for pole pass sidebands.
Noisy bearings	Lack of grease or wrong grease	Regrease per M-OM-03 section 5.
Overload relay trips	Overload, wrong relay setting, single phasing	Check current per phase and relay setting.
Motor does not start	No supply, blown fuse, motor or load locked	Check supply and that the shaft turns freely after isolation.

4. Troubleshooting motor overheating

Use this procedure when the winding temperature reaches the warning limit of **130 °C**, or when the motor has tripped on the winding temperature alarm of **145 °C**.

1 **Do not reset and restart repeatedly.** A motor that tripped on temperature must cool down before restart. Repeated restarts add starting heat and can destroy the winding.

2 **Confirm the measurement.** Compare the three PT100 readings. One sensor reading much higher than the others may be a sensor or wiring fault. Check the frame temperature with an infrared thermometer: at full load the frame is typically 40 to 60 °C above ambient.

3 **Check the load current.** Measure the current in all three phases and compare with the rated current on the nameplate. A current above rated current means the motor is overloaded: check the driven machine (for a conveyor: jammed rollers, belt tension, product build-up).

4 **Check the supply.** Measure the three phase-to-phase voltages. Calculate the voltage imbalance; it must not exceed 1%. A missing phase or a high-resistance connection gives one phase with high or zero current.

5 **Check the cooling.** Inspect the fan and fan cover, clean the cooling fins and make sure there is free space behind the fan cover. Check that the motor is not drawing in hot air.

6 **Check the duty.** Review the start/stop history. More than 3 starts per hour overheats the motor.

7 **Check the bearings.** A failing bearing adds friction heat and noise, mainly at the drive end.

8 **Check the winding insulation.** If the motor has been running above 145 °C, a qualified electrical technician must measure the insulation resistance (M-ETP-05) before the motor is returned to service.

9 **Record the findings** in a maintenance ticket with the measured currents, voltages and temperatures.

5. Bearing replacement

1 Isolate the motor according to SOP-LOTO-001 and uncouple it from the driven machine.

2 Remove the fan cover and the fan. Mark the position of the end shields.

3 Remove the end shields and withdraw the rotor carefully, supporting it so that it does not touch the stator winding.

4 Pull the bearings off with a puller acting on the inner ring.

5 Heat the new bearings with an induction heater to a maximum of **110 °C** and slide them onto the shaft. Never hammer a bearing onto the shaft.

6 Fill the bearing and the grease chamber about one half with fresh grease of the specified type.

7 Reassemble the motor, check that the rotor turns freely and measure the insulation resistance before connecting the supply.

6. Checking for broken rotor bars

Broken rotor bars cause current fluctuations at twice slip frequency and vibration at running speed with sidebands. Typical signs are a cyclic "beating" noise under load and a slow pulsation of the ammeter needle. Confirm with motor current signature analysis; a sideband level less than 45 dB below the supply frequency peak indicates a broken bar.